

Symptom-Based Classification of Common Syndromes Using Machine Learning: A Review

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ABSTRACT

Symptom-based disease classification has emerged as a critical application of machine learning (ML) and artificial intelligence (AI) in modern healthcare. With the growing availability of electronic health records, patient-reported outcomes, and conversational health systems, automated interpretation of symptoms offers an efficient approach for early diagnosis and clinical decision support. This review systematically analyzes recent advancements in symptom-based classification of common syndromes using traditional machine learning, deep learning, natural language processing (NLP), and large language models (LLMs). The study highlights methods that process structured symptom vectors as well as unstructured free-text symptom descriptions obtained from chatbots and clinical narratives. A comprehensive comparison of twenty recent studies published between 2024 and 2025 is presented, focusing on employed methodologies, key advantages, and reported limitations. Furthermore, the review synthesizes major research findings, identifies open challenges such as data sparsity, explainability, and clinical reliability, and discusses future research directions. This paper aims to serve as a consolidated reference for researchers and practitioners working on intelligent symptom-based disease prediction systems and to guide future development of robust, interpretable, and clinically deployable models.

Keywords: Symptom Classification, Machine Learning, Disease Prediction, Natural Language Processing, Healthcare AI.

Introduction

The rapid digitalization of healthcare systems has led to an unprecedented growth in patient-generated health data, particularly symptom descriptions

captured through electronic health records, mobile health applications, and conversational agents. Symptoms often represent the earliest observable indicators of disease and play a crucial role in clinical

diagnosis. However, manual interpretation of symptom patterns is time-consuming and highly dependent on clinician expertise. As a result, symptom-based classification using machine learning has gained considerable attention as a scalable and cost-effective diagnostic support mechanism [1,2].

The primary aim of symptom-based syndrome classification is to automatically map a set of observed symptoms to one or more probable diseases or syndromes. Early research relied heavily on rule-based expert systems and statistical models, which suffered from limited adaptability and scalability. With the emergence of machine learning, models such as decision trees, support vector machines, and ensemble learners enabled more accurate classification by learning symptom-disease relationships from data [3,12]. These methods demonstrated improved performance but often required structured symptom inputs and struggled with incomplete or noisy data.

Recent advancements in deep learning and NLP have significantly transformed this domain. Free-text symptom descriptions, commonly provided by patients, can now be analyzed using word embeddings, transformer-based architectures, and attention mechanisms [1,2,9]. Moreover, the integration of large language models has enabled conversational diagnosis and differential reasoning across multiple diseases, expanding the scope of symptom-based systems beyond simple classification [6,8,10]. These models can capture contextual and semantic nuances in symptom narratives, making them suitable for real-world clinical settings.

The objective of this review is threefold. First, it systematically surveys recent literature on symptom-based disease and syndrome classification using ML and AI techniques. Second, it compares methodologies, advantages, and limitations of state-

of-the-art approaches through a structured literature table. Third, it synthesizes research findings, identifies persistent challenges, and outlines future research directions. By doing so, this paper provides a holistic overview of current trends and gaps in symptom-based classification systems, thereby supporting researchers in developing more accurate, explainable, and clinically reliable diagnostic tools [4–8].

LITERATURE STUDY

The existing literature on symptom-based disease classification demonstrates a clear transition from traditional machine learning to deep learning and LLM-driven approaches. Early works focused on structured symptom datasets and employed classifiers such as SVMs, decision trees, and association-rule-based recommender systems [3,12,14]. These approaches offered simplicity and interpretability but were limited in handling free-text inputs and complex symptom interactions. More recent studies have leveraged NLP pipelines, BERT-based architectures, and GAN-based data augmentation to improve robustness and generalization, particularly in scenarios with limited labeled data [1,9,17].

A second stream of research explores conversational agents and LLM-powered diagnostic systems. Chatbot-based frameworks and differential diagnosis systems demonstrate promising results in extracting symptoms, reasoning across diseases, and assisting clinicians in decision-making [4,6,8,18]. However, concerns related to hallucination, misinformation, and lack of clinical explainability remain significant. Table 1 summarizes twenty representative studies, highlighting their methodologies, advantages, and limitations, thereby offering a concise comparative view of recent advancements [10–20].

TABLE I COMPARATIVE ANALYSIS

Ref No.	Year – Publication	Methods	Advantage	Limitation / Future Work
[1]	2025 – IJACSA	DL + NLP	Handles free-text symptoms	Limited explainability
[2]	2025 – JEEEMI	NLP + ML	Multi-disease prediction	Needs clinical validation
[3]	2025 – THI	SVM, DT	Simple implementation	Low scalability
[4]	2025 – IEEE JBHI	Conversational AI	Rare disease focus	Data dependency
[5]	2025 – AI in Agriculture	Multimodal DL	Robust fusion	Domain-specific
[6]	2025 – npj Health	LLM dual inference	Explainable reasoning	Computational cost
[7]	2025 – SLAS Tech	NLP + LLM	Biomarker discovery	Indirect diagnosis
[8]	2025 – AI	LLM analysis	Broad symptom coverage	Risk of hallucination
[9]	2025 – Sci. Reports	BERT + GAN	Data augmentation	Training complexity
[10]	2025 – IEEE Access	LLM evaluation	Diagnostic insight	Misinformation risk
[11]	2025 – JHS	LLM analysis	Error identification	Limited scope
[12]	2024 – DAJ	Association rules	Handles incomplete data	Rule explosion
[13]	2024 – IEEE Access	MediGPT	Multi-model comparison	Ethical concerns
[14]	2024 – IJIT	ML ensemble	Comparative analysis	Static datasets
[15]	2024 – CMI	ChatGPT NLP	Real-world data	Bias issues
[16]	2024 – medRxiv	DxGPT	Rare disease diagnosis	Needs regulation
[17]	2024 – Sci. Reports	LM optimization	High accuracy	Interpretability
[18]	2024 – Sci. Reports	LLM chatbot	Chronic disease aid	Long-term reliability
[19]	2024 – JBI	Text mining	Language resources	Localization needed
[20]	2024 – Frontiers Med.	LLM + KG	Knowledge integration	Maintenance overhead

RESEARCH FINDING

Analysis of the reviewed literature reveals several important findings regarding symptom-based classification systems. First, the shift from structured symptom checklists to free-text symptom narratives has significantly enhanced usability and patient engagement [1,2,15]. NLP-driven models, particularly transformer-based architectures, demonstrate superior performance in capturing semantic relationships among symptoms compared to traditional bag-of-words approaches [9,17]. This has resulted in improved classification accuracy across multi-disease scenarios.

Second, hybrid systems that combine machine learning with knowledge-driven components, such as knowledge graphs and rule-based reasoning, offer improved robustness and contextual understanding [12,20]. These systems are particularly effective when dealing with incomplete or ambiguous symptom sets, which are common in real-world clinical settings. Third, LLM-based diagnostic frameworks exhibit strong potential in conversational diagnosis and differential reasoning, enabling multi-turn symptom clarification and explanation generation [6,8,18]. However, their performance is highly sensitive to prompt design, training data quality, and domain adaptation.

Another key finding is the growing emphasis on explainability and trustworthiness. Several studies incorporate attention mechanisms, rule extraction, or dual-inference strategies to improve transparency and clinician trust [6,11]. Despite these efforts, explainability remains an open issue, especially for deep neural and LLM-based models. Finally, most studies report strong experimental performance but lack large-scale clinical validation, highlighting a gap between research prototypes and deployable healthcare solutions [10,13,16].

CHALLENGES

Despite rapid progress in symptom-based disease and syndrome classification using machine learning, several fundamental challenges continue to limit real-world adoption and clinical reliability. These challenges arise from the nature of symptom data, model design constraints, ethical considerations, and integration barriers within healthcare systems. Addressing these issues is essential for developing safe, accurate, and trustworthy diagnostic support systems.

1. Data Quality, Noise, and Symptom Ambiguity

One of the most critical challenges in symptom-based classification is the inherent quality and ambiguity of symptom data. Symptoms are often self-reported by patients and expressed in subjective, vague, or inconsistent language. Different individuals may describe the same symptom using varied terminology, intensity levels, or contextual expressions. Additionally, real-world datasets frequently contain missing symptoms, contradictory entries, or recording errors. These issues introduce noise and bias into training data, leading to reduced model accuracy and unstable predictions. Although techniques such as data preprocessing, symptom normalization, and augmentation have been explored, they are not sufficient to fully resolve semantic ambiguity and sparsity. As a result, models may misclassify diseases with overlapping symptom profiles, especially in early-stage or atypical presentations.

2. Limited Interpretability and Explainability of Models

Another major challenge lies in the lack of interpretability of advanced machine learning and deep learning models. While deep neural networks and large language models demonstrate high predictive performance, they often function as “black boxes,” providing limited insight into how specific symptoms contribute to a diagnosis. In clinical environments, healthcare professionals require transparent reasoning to trust and validate automated recommendations. The absence of explainability not only reduces clinician confidence but also complicates error analysis and accountability. Although attention mechanisms, feature importance analysis, and explainable AI frameworks have been proposed, these solutions remain partial and are not yet standardized across symptom-based diagnostic systems.

3. Generalization Across Populations and Healthcare Settings

Generalization remains a persistent challenge due to demographic, linguistic, and regional variations in symptom expression. Models trained on data from a specific population or healthcare system may perform poorly when deployed in different geographic regions or cultural contexts. Language differences, healthcare-seeking behavior, and prevalence of diseases significantly influence symptom patterns. Furthermore, many datasets are collected under controlled or limited clinical conditions, which do not reflect real-world diversity. This lack of generalizability restricts large-scale deployment and may introduce unfair biases, particularly affecting underrepresented populations.

4. Ethical, Safety, and Reliability Concerns

The use of AI-driven symptom-based classification systems raises serious ethical and safety concerns. Large language models and conversational diagnostic tools may generate hallucinated or misleading outputs that appear clinically plausible but are factually incorrect. Such misinformation can lead to delayed treatment, patient anxiety, or inappropriate self-

medication. Moreover, issues related to data privacy, informed consent, and accountability remain unresolved. Without clear regulatory frameworks and validation standards, deploying these systems in healthcare settings poses legal and ethical risks. Ensuring patient safety while maintaining innovation is a delicate balance that must be carefully managed.

5. Integration with Clinical Workflows and Regulatory Compliance

Finally, integrating symptom-based ML systems into existing clinical workflows presents practical and regulatory challenges. Healthcare environments are complex, time-sensitive, and governed by strict standards. Diagnostic support systems must seamlessly interact with electronic health records, decision-support tools, and clinician workflows without increasing cognitive or administrative burden. Additionally, regulatory approval, continuous monitoring, and post-deployment evaluation are resource-intensive processes. Many current models remain research prototypes and lack the robustness, validation, and compliance required for routine clinical use.

CONCLUSION AND FUTURE WORK

This review presented a comprehensive analysis of symptom-based classification of common syndromes using machine learning and AI. Recent advances in NLP, deep learning, and LLMs have significantly improved the ability to interpret free-text symptoms and support multi-disease diagnosis. However, challenges related to data quality, interpretability, and clinical reliability persist.

Future research should focus on hybrid explainable models, large-scale clinical validation, and ethical deployment frameworks. Emphasis on multilingual datasets, federated learning, and clinician-in-the-loop systems can further enhance trust and applicability of symptom-based diagnostic tools in real-world healthcare environments.

References

- [1]. S. S. Al-qarni and A. Algarni, "Disease Prediction from Symptom Descriptions Using Deep Learning and NLP Technique," *International Journal of Advanced Computer Science and Applications*, vol. 16, no. 5, pp. 416–426, 2025, doi: 10.14569/IJACSA.2025.0160541.
- [2]. D. Vyas, M. Shah, H. Kantawala, B. Patel, T. Patel, and J. Enamala, "SympTextML: Leveraging Natural Language Symptom Descriptions for Accurate Multi-Disease Prediction," *Journal of Electronics, Electromedical Engineering, and Medical Informatics*, vol. 7, no. 3, pp. 911–924, 2025, doi: 10.35882/jeeemi.v7i3.946.
- [3]. A. Chandel, "Healthchare Chatbot Using SVM & Decision Tree," *Trends in Health Informatics*, vol. 2, no. 1, pp. 10–17, 2025.
- [4]. J. Yang, L. Shu, H. Duan, and H. Li, "RDguru: A Conversational Intelligent Agent for Rare Diseases," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 9, pp. 6366–6378, 2025, doi: 10.1109/JBHI.2024.3464555.
- [5]. G. Yang et al., "TGFN-SD: A text-guided multimodal fusion network for swine disease diagnosis," *Artificial Intelligence in Agriculture*, vol. 15, no. 2, pp. 266–279, 2025, doi: 10.1016/j.aiia.2025.03.002.
- [6]. S. Zhou et al., "Explainable differential diagnosis with dual-inference large language models," *npj Health Systems*, vol. 2, no. 1, pp. 1–9, 2025, doi: 10.1038/s44401-025-00015-6.
- [7]. J. T. Song, J. J. Huang, and R. L. Liu, "Integrating NLP and LLMs to discover biomarkers and mechanisms in Alzheimer's disease," *SLAS Technology*, vol. 31, no. February, pp. 0–9, 2025, doi: 10.1016/j.slant.2025.100257.
- [8]. G. K. Gupta, A. Singh, S. V. Manikandan, and A. Ehtesham, "Digital Diagnostics: The

- Potential of Large Language Models in Recognizing Symptoms of Common Illnesses,” AI (Switzerland), vol. 6, no. 1, pp. 1–17, 2025, doi: 10.3390/ai6010013.
- [9]. X. Chen and Y. Du, “Enhancing medical text classification with GAN-based data augmentation and multi-task learning in BERT,” *Scientific Reports*, vol. 15, no. 1, pp. 1–13, 2025, doi: 10.1038/s41598-025-98281-9.
- [10]. I. Almubark, “Exploring the Impact of Large Language Models on Disease Diagnosis,” *IEEE Access*, vol. 13, no. December 2024, pp. 8225–8238, 2025, doi: 10.1109/ACCESS.2025.3527025.
- [11]. G. E. Sayegh, D. Ring, and P. Jayakumar, “Potential misinformation in large language model descriptions of upper extremity diseases,” *Journal of Hand Surgery: European Volume*, vol. 50, no. 3, pp. 411–414, 2025, doi: 10.1177/17531934241268975.
- [12]. K. N. Singh and J. K. Mantri, “An intelligent recommender system using machine learning association rules and rough set for disease prediction from incomplete symptom set,” *Decision Analytics Journal*, vol. 11, no. April, p. 100468, 2024, doi: 10.1016/j.dajour.2024.100468.
- [13]. M. Abu Tareq Rony, M. Shariful Islam, T. Sultan, S. Alshathri, and W. El-Shafai, “MediGPT: Exploring Potentials of Conventional and Large Language Models on Medical Data,” *IEEE Access*, vol. 12, no. August, pp. 103473–103487, 2024, doi: 10.1109/ACCESS.2024.3428918.
- [14]. A. Das, D. Choudhury, and A. Sen, “A collaborative empirical analysis on machine learning based disease prediction in health care system,” *International Journal of Information Technology (Singapore)*, vol. 16, no. 1, pp. 261–270, 2024, doi: 10.1007/s41870-023-01556-5.
- [15]. W. I. Wei, C. L. K. Leung, A. Tang, E. B. McNeil, S. Y. S. Wong, and K. O. Kwok, “Extracting symptoms from free-text responses using ChatGPT among COVID-19 cases in Hong Kong,” *Clinical Microbiology and Infection*, vol. 30, no. 1, pp. 142.e1–142.e3, 2024, doi: 10.1016/j.cmi.2023.11.002.
- [16]. J. do Olmo, J. Logroño, C. Mascías, M. Martínez, and J. Isla, “Assessing DxGPT: Diagnosing Rare Diseases with Various Large Language Models,” *med*, pp. 1–13, May 2024, doi: 10.1101/2024.05.08.24307062.
- [17]. E. Hassan, T. Abd El-Hafeez, and M. Y. Shams, “Optimizing classification of diseases through language model analysis of symptoms,” *Scientific Reports*, vol. 14, no. 1, pp. 1–24, 2024, doi: 10.1038/s41598-024-51615-5.
- [18]. S. Zhang and J. Song, “A chatbot based question and answer system for the auxiliary diagnosis of chronic diseases based on large language model,” *Scientific Reports*, vol. 14, no. 1, pp. 1–14, 2024, doi: 10.1038/s41598-024-67429-4.
- [19]. H. Zong et al., “Advancing Chinese biomedical text mining with community challenges,” *Journal of Biomedical Informatics*, vol. 157, no. August, 2024, doi: 10.1016/j.jbi.2024.104716.
- [20]. Y. Duan et al., “Research on a traditional Chinese medicine case-based question-answering system integrating large language models and knowledge graphs,” *Frontiers in Medicine*, vol. 11, no. January, 2024, doi: 10.3389/fmed.2024.1512329.